Culverizer

A 3 lb horizontal-spinner combat robot



Type / weight

Horizontal spinner / 3 lb / Beetleweight

Version

v4.2

Win record (v4.2)

18-15 / 54.5% winrate

Competitions (v4.2)

7 tournaments

Energy stored

700 J

My role

Lead Engineer: design, manufacturing, electronics, & testing
Competition strategy and drive coach

Design approach

I designed Culverizer around an offense-first strategy, using a heavy, hardened steel blade to destroy opponents' wheels and batter their weapon assemblies. Carbon-fiber plates provide chassis rigidity and support the weapon assembly, while the 3D-printed 95A TPU body flexes to absorb and disperse impact energy. The TPU serves as sacrificial armor, taking damage to protect the internal electronics.

Drivetrain

Brushless motors propel Culverizer at 12 mph. TPU shock mounts let the motors deflect under impact, directing energy away from the gearboxes to extend their lifespan. I made custom molds for 2.1-inch wheels with impact-absorbing TPU cores and 45A urethane treads. The urethane adheres to the TPU and provides traction and acceleration; 1/32-inch spring-steel "spiked cleats" further improve traction and control. The compliant cores also help protect the wheels and gearmotors from impact.

Weapon Assembly

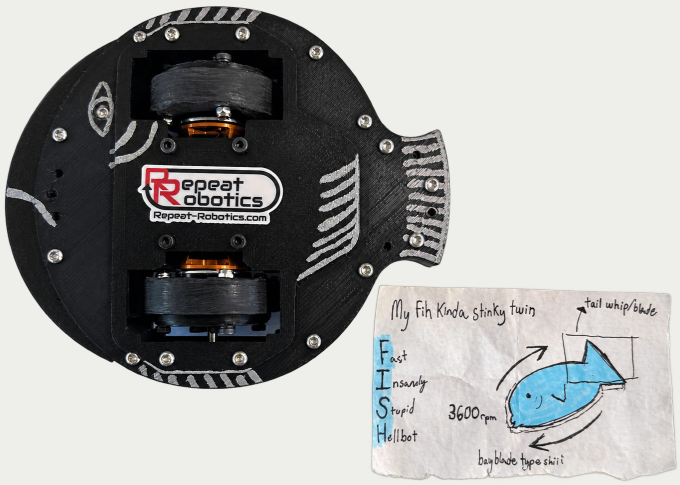
A 600 kV motor at 25.2 V drives the blade through a 6:5 belt reduction, reaching a measured 220 mph and storing 700 J of energy. The flat timing belt allows the assembly to slip on impact, reducing shock transferred to the motor. A spline pattern transfers torque to the blade.

Competition Results

I built three working copies of Culverizer, configured to counter different opponent build styles. Across seven competitions, v4.2 earned an 18-15 record (54.5% win rate), including two third-place finishes.

FIH

A 1 lb plastic antweight built in 48 hours



Type / weight

Melty brain / 1 lb / Plastic antweight

Version

Buildathon

Win record (Buildathon)

4-2 / 66.7% winrate

Competitions (Buildathon)

2 Buildathon competitions

Energy stored

Unknown

FIH with its original fish concept sketch.

My role

Team of two; I led much of the design, focusing on drivetrain and chassis.
Kavan Ramaswamy designed the wheel mounting and handled assembly, soldering, and wheel manufacturing.

Design approach

At Railside Robotics' 48-hour Buildathon over Labor Day weekend in September 2026, each team received 1 kg of PLA to build a one-pound plastic antweight. Our two-person team was given a fish-shaped sketch as our assigned design. The challenge was to interpret it and match the drawing as closely and thoughtfully as possible. Because the sketch resembled a melty brain, we chose to make it function as one, competing after 24 hours and again at 48 hours. I led much of the design, focusing on the drivetrain and chassis. We used Fusion 360 because Kavan was more proficient in it; although I was more familiar with SolidWorks, I adapted quickly so we could collaborate. Kavan designed the wheel mounting and handled assembly, soldering, and wheel manufacturing.

Building and driving a melty brain

A typical melty brain uses a controller to measure rotation and pulse the motors for translation. Without that controller, I manually varied the relative speed of the two drive motors while driving. The approach required estimation, but produced useful translation and let me line up shots. Balance was a key challenge. We designed for balance in CAD, then added screws as ballast to adjust the center of gravity. We also designed the PLA motor mounts with enough material around the motors to withstand heat when they stalled; the mounts held up. To work within the tight deadline, we designed chassis components as flat pieces rather than tall parts. Printing with less height let us iterate faster and produce spare parts in time for competition.

FIH (continued)

Testing and iteration

After the 24-hour competition exposed chassis structural issues, we added material along the weapon-impact load path. The revised chassis held together well in the second competition. In the second competition, foam wheels failed and screws backed out. We reduced wheel speed for the next match, but could not fully resolve the issue during the event. In a future revision, I would use a more robust wheel material, improve screw retention, and consider lower-kV motors: the available motors were operated at about 30%, and the robot experienced heating and stalls.

Competition Results

FIH went 3-0 and won the 24-hour competition. At 48 hours, it went 1-2; both losses involved motor stalls and a depleted battery. Across both competitions, the combined record was 4-2.